



Prediction of drug protein interactions based on variable scale characteristic pyramid convolution network

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ABSTRACT

Motivation: In the process of drug screening, it is significant to improve the accuracy of drug-target binding affinity prediction. A multilayer convolutional neural network is one of the most popular existing methods for predicting affinity based on deep learning. It uses multiple convolution layers to extract features from the simplified molecular input system (SMILES) strings of the compounds and amino acid sequences of proteins and then performs affinity prediction analysis. However, the semantic information contained in low-level features can gradually be lost due to the increasing network depth, which affects the prediction performance.

Result: We propose a novel method called the Pyramid Network Convolution Drug-Target Binding Affinity (PCNN-DTA) method for drug-target binding affinity prediction. The PCNN-DTA method, which is based on a feature pyramid network (FPN), fuses the features extracted from each layer of a multilayer convolution network to retain more low-level feature information, thus improving the prediction accuracy. PCNN-DTA is compared with other typical algorithms on three benchmark datasets, namely, the KIBA, Davis, and Binding DB datasets. Experimental results show that the PCNN-DTA method is superior to existing regression prediction methods using convolutional neural networks, which further demonstrates its effectiveness.

1. Introduction

A crucial step in drug discovery is identifying ligand affinity with a target, which can be reliably conducted via wet-lab (in vitro) experiments. However, wet-lab experiments suffer from labor-intensive, high cost and long cycle [1]. To relieve these bottlenecks, the approach of drug-target binding affinity (DTA) prediction problem with computational methods has attracted considerable attention in recent years due to their convenience and low cost.

According to source of the drugs and targets features of these computational methods, they can be divided into two categories: structural approaches and structural-free approaches [2]. Among various structure-based methods, no matter the simple scoring functions, the molecular dynamics (MD) simulations or the deep learning-based methods, all of them require the three-dimensional (3D) structure information of drugs and targets. Although these methods have achieved attractive performance in DTA prediction problem and can provide reasonable interpretability, their application scope is limited due to difficulties to obtain exact 3D structure of targets [3–6].

Recently, many researchers hoped to use structural-free of drug and

target to expand the coverage of DTA computational methods. In early structural-free methods, traditional machine learning has been employed, such as Kron-SVM [7] and BRDTI [8]. The performances of these methods rely too much on hand-crafted features. With the advancement of deep learning methods and growth of biological databases, numerous deep-learning-based structural-free methodologies are emerging in the field of drug discovery. These deep learning based DTA models employ CNN, RNN, GNN and Transformer to learn more robust high-level representations from raw chemical structures of drugs and amino acid sequences of target [9–15]. The representative method in these works is DeepDTA [14], which employed the one-dimensional CNN as the feature extractors for the protein sequences and drug simplified molecular input system (SMILES). Lee et al. also utilized the CNN to extract local residue patterns of protein sequences [16]. The results from their works demonstrate that the learned features performed better than other protein descriptors. In [17], drug SMILES and protein sequences were sent into two CNN modules to extract local features. To better learn the interaction information between the two extracted features, researchers proposed a mutual learning mechanism for the interpretable drug-target interaction prediction method. Gu et al.

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proposed a co-regularized variational autoencoders (Co-VAE) to predicted binding affinity which also employed the CNN as backbone network [18]. Zi duo Yang et al [19] proposed ML-DTI method by combining mutual learning with deep convolution network. This method can realize mutual learning of feature extraction of drug small molecules and protein sub molecules that are independent of each other in previous methods.

The above DTA prediction methods based on deep learning follow similar paradigm. First, the protein and drug molecular representation sequences as the inputs for the CNN to extract features. Secondly, the extracted features were fused and passed through several dense layers to predict the binding affinity. Overall, many CNN-based methods have been developed and achieved remarkable performance on various datasets. However, at least one problem has not been well addressed for CNN-based methods in DTA prediction. These deep learning methods fail to consider the information of different CNN layers, which will lead to partial information loss. In the field of Computer Vision, many studies have proved that different levels of CNN networks contain different information. Existing DTA methods pay much attention to final output information of CNNs, and in this work, we use different level of information from CNNs as supplement.

The motivation of this paper is to effectively integrate all the information of CNN at different layers for more accurate DTA prediction. To this end, we propose a pyramid CNN for DTA (PCNN-DTA), which jointly learns high-level semantic and the low-level detailed

information. Specifically, our proposed PCNN-DTA method contains two parts: two feature pyramid networks based on a CNN for extracting features and a regression block to predict the binding affinities based on the learned representations. The main contribution can be summarized as follows:

1. We propose a novel pyramid CNN structure for DTA prediction by building a Feature Pyramid Network as our feature extractor, which combines all the features of convolutional layers via a top-down pathway and skip-connections.
2. We design a multi-head self-attention mechanism (MHSA) to fuse the features from the pyramid CNN, to improve the performance of DTA prediction.

2. Problem formulation

In our method, DTA prediction is regarded as a regression problem. Let $X^{Train} = \{(c^{(i)}, p^{(i)})\}_{i=1}^N$ denote the training samples, where $c^{(i)}$ indicates the i th compound SMILES string, $p^{(i)}$ indicates the i th protein sequence, and N represents the number of available pairs of compounds and proteins. Let $Y^{Train} = \{y^{(i)}\}_{i=1}^N$ denote the corresponding labels of each C–P pair. Similarly, test samples are represented as X^{Test} . The goal of our method is to learn a function $f(\cdot)$ to predict the binding affinity values.

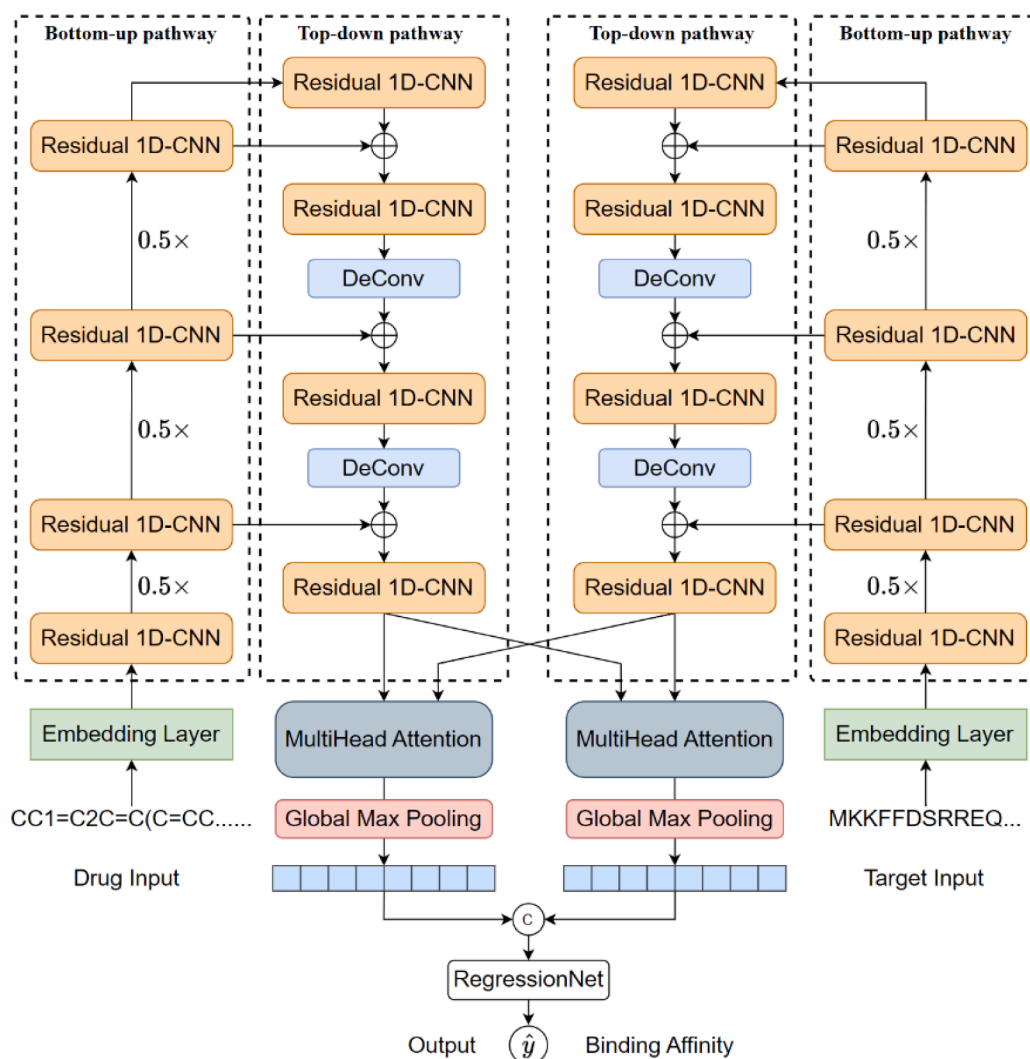


Fig. 1. Overall flow chart of the algorithm.

3. Method

In this study, the proposed PCNN-DTA method consists of two embedding layers for encoding the drug SMILES and protein amino acid sequence respectively, two feature pyramid networks for extracting high-level abstract features from the embeddings of drug and target, a feature fusion module for integrating the drug feature and target feature, and a regression block for binding affinity prediction based on the fusion features (the overall structure diagram is shown in Fig. 1). Given the input of a compound SMILES string and protein sequence $(c^{(i)}, p^{(i)})$ pair, the model first converts the inputs into two low-dimensional matrices using the embedding layers. Then, the matrices of compounds and proteins are fed into two feature pyramid networks with the similar structure. Each feature pyramid networks contains a bottom-up pathway and a top-down pathway for extracting different level characteristics of proteins and compounds. After that, the feature vectors of the compounds and proteins are sent to the feature fusion module. This fusion module uses a multi-head attention mechanism to generate the interaction features between the proteins and drugs. Finally, all interaction features are merged and transmitted to a regression block composed of four fully connected layers for the final binding affinity prediction and analysis.

3.1. Embedding layer

In this work, SMILES sequences and amino acid sequences were used to express compound molecules and proteins. A SMILES sequence is a specification that uses ASCII strings to clearly describe the molecular structure. Moreover, compound and protein inputs were converted using integer/label encoding. The fixed length of the compounds and proteins were set to 100 and 1000, and those less than 100 and 1000 were padded with zeros to have the same length.

After being expressed by SMILES and amino acids, we employed an embedding layer to convert each character into an L -dimensional vector representation for the protein P and compound C , where $P \in R^{|P| \times V_p}$, $C \in R^{|C| \times V_c}$, V_p represents the size of the amino acid vocabulary and V_c represents the size of the SMILES vocabulary.

3.2. Feature pyramid networks

Fig. 2 illustrates the architecture of the feature pyramid network for drug SMILES or protein sequence. The feature pyramid network can learn the information from all levels of CNN. It consists of a bottom-up pathway, a top-down pathway, and lateral connections. The detailed description of feature pyramid network is given as follows.

Bottom-up pathway. In the bottom-up pathway, we stacked four residual 1D-CNN to generate feature maps for the embedding of drug SMILES and protein sequence respectively. The stacked residual 1D-CNN module is an architecture composed of convolutional layers with an activation function with skip-connection. The convolutional layer is the major unit of residual 1D-CNN module, which applies convolutional filters to the input embedding matrix M (P or C). Each filter is moved incrementally through the input embedding as a sliding window (with window size k) to obtain the compositional representation of k neighboring terms. Suppose a convolutional layer has F filters, and the CNN layer produces an output matrix $M^O \in R^{|M| \times F}$ with $O(F \times k \times L)$ parameters. The residual 1D-CNN layer can be formulated as

$$M^h = M^{h-1} + \text{conv}(M^{h-1}) \\ = \text{Gelu} \left(\sum_{F_i} \sum_{k_i} \sum_{L_i} M(k_i) * F_i(L_i) \right) \quad (1)$$

where M^{h-1} represents the output matrix of the $(h-1)$ th residual 1D-CNN layer. Note that $M^0 = M$ represents the input matrix obtained directly from the embedding layer.

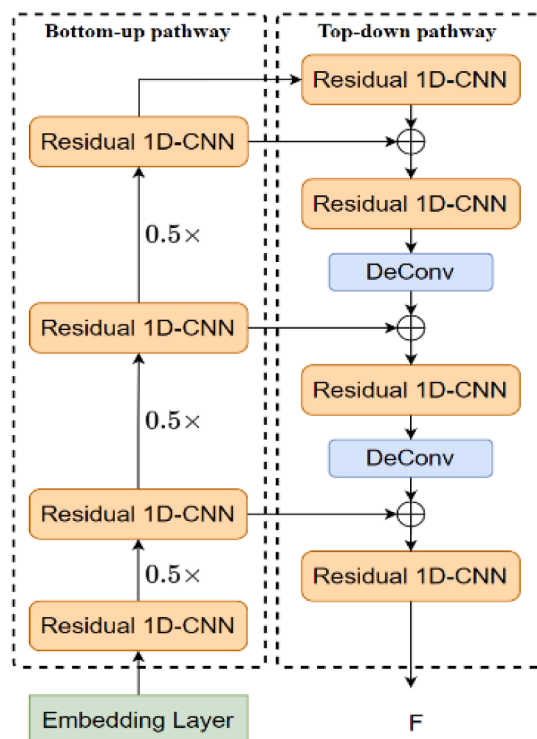


Fig. 2. FPN structure diagram.

To obtain higher-level representations, we stacked multiple convolutional layers hierarchically.

In the residual 1D-CNN module, the consecutive convolutional layers allow us to acquire higher-level abstractions of the input. The higher the layer, the higher the spatial resolution for localization of the features they output, but the output of higher layers also has lower semantic information for recognition.

Specifically, we denote the output of each residual 1D-CNN layer for drug and target as $\{D1, D2, D3, D4\}$ and $\{T1, T2, T3, T4\}$, and note that they have strides of $\{1, 2, 4, 8\}$ sequence with respect to the input sequence.

Top-down pathway. The bottom-up pathway conveys the strong semantically features for deep CNN layers but ignore the shallow ones. For the purpose of using the extracted shallow information, we employed the top-down pathway to enhance the representation. Specifically, the bottom-up pathway consists of a max-pooling layer followed by residual 1D-CNN to convey higher abstracted features. Symmetrical with the top-down pathway, downsampled features are the increased by deconvolution then merged with the corresponding upper features from the bottom-up pathway via lateral connections, as shown in Fig. 2.

Considering that the more semantically stronger feature maps generated by our bottom-up pathway are coarser-resolution, we upsample the spatial resolution by a deconvolution layer by stride of 2. The deconvolution layer similar to the convolution layer of bottom-up pathway which also followed by a layer normalization and a GELU. The deconvolution operation can be formulated as:

$$F = G(X, \varphi) \quad (2)$$

where φ is the parameters of deconvolution operation, X and F are the input and output of deconvolution respectively. In our method, the kernel of all deconvolutions is set to 1×3 . The upsampled feature maps via deconvolution layer is then merged with the corresponding bottom-up map by element-wise addition. In addition, to reduce the up-sampling aliasing effect, the residual 1D convolution is subsequently applied to the output of the element-by-element addition operation.

Finally, the drug and target features are generated as F_D, F_T .

3.3. Fusion mechanism

In this study, Multi-Head Attention (MHA) was introduced in the feature fusion block. In MHA, the linear transformation layers first process the Q, X, and V of each head, as shown in Eqs. (3), (4), and (5), where W_Q, W_K, W_V are the linear parameter matrices of Q, K, and V, respectively. The scaled dot product attention is calculated as shown in Eq. (6). The input to the fusion block comes from the output pairs of each feature pyramid network, and the input pairs of the fusion block are processed using multi-attention. The attention scores of all heads are concatenated as the input to acquire the multi-head attention, which is shown in Eq. (7), where W_O is the linear parameter matrix.

$$E_P^{Q'} = E^P \times W_i^Q \quad (3)$$

$$E_P^{K'} = E^P \times W_i^K \quad (4)$$

$$E_P^{V'} = E^P \times W_i^V \quad (5)$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{E^P (E^C)^T}{\sqrt{d_k}}\right) E^C \quad (6)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (7)$$

where

$$\text{head}_i = \text{Attention}(E^P W_i^Q, E^C W_i^K, E^C W_i^V) \quad (8)$$

For the equations above, $W_i^Q \in R^{d_{\text{model}} \times d_q}$, $W_i^K \in R^{d_{\text{model}} \times d_k}$, and $W_i^V \in R^{d_{\text{model}} \times d_v}$ are parameter matrices, d_q, d_v , and d_k denote the Query vector, Value vector and Key vector dimensions, respectively, and h denotes the number of heads.

As shown in Fig. 2 and Eq. (2), the output F_D, F_T of drug feature pyramid network and target feature pyramid network are regarded as queries, keys and values matrices respectively and are fed into two multi-head attention. Finally, the global representations G_D, G_T for drug and target are generated, which consider multi-level features from CNN and interaction information between drug and target by multi-head attention.

$$G_D = \text{MultiHead}(F_D, F_D, F_T) \quad (9)$$

$$G_T = \text{MultiHead}(F_T, F_T, F_D) \quad (10)$$

3.4. Binding affinity prediction

The obtained global features are followed by two global max pooling to reduce the dimensional and then concatenated together and sent to the a four-layer fully connected network to predict the binding affinity.

3.5. Loss function

Mean Squared Error (MSE) was selected as the loss function.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (P_i - Y_i)^2 \quad (11)$$

MSE represents the error between the estimated value and the actual value, and the parameters can be adjusted according to MSE. In the formula 11, P and Y represent estimated value and actual value respectively.

4. Model validation

4.1. Evaluation metrics

The concordance index, mean squared error (MSE), Pearson correlation coefficient (R) and r_m^2 index were used to evaluate the model.

$$CI = \frac{1}{Z} \sum_{\delta_x > \delta_y} h(b_x - b_y) \quad (12)$$

$$\text{where } h(m) = \begin{cases} 1 & \text{if } m > 0 \\ 0.5 & \text{if } m = 0 \\ 0 & \text{if } m < 0 \end{cases}$$

$$R = \frac{\text{cov}(p, y)}{\sigma(p)\sigma(y)} \quad (13)$$

$$r_m^2 = r^{2*} \left(1 - \sqrt{r^2 - r_0^2}\right) \quad (14)$$

The C-index (CI) reflects the conformity between the predicted results of the model and the actual observation results to evaluate the accuracy of the model, which can be calculated as in (12). The value range of C-index is 0 to 1. When the C-index is less than 0.5, the prediction accuracy of the model is not as good as that of the random model. If C-index = 1, the prediction result of the model is completely consistent with the actual result.

The MSE can be calculated using Eq. (11). R is Pearson coefficient, which is used to measure the correlation between two vectors. In this paper, R is used to measure the correlation between the prediction vector and the actual vector and is calculated as in (13). The r_m^2 index is calculated as Eq. (14), where r^2 and r_0^2 are the squared correlation coefficients with and 0 without intercepts, respectively.

We evaluated our method under two settings: random split setting and cold-start setting. Fig. 3, Fig. 4, Fig. 5, Fig. 6 shows the distributions between the training, validation and testing sets under the random setting and cold-start setting for the Davis and KIBA datasets.

4.2. The performance of PCNN-DTA

We evaluated PCNNDTA on two drug–target affinity predictions publicly available datasets: Davis [20] and KIBA [21] datasets. We compared with the DeepDTA [14], GraphDTA [15], and ML-DTI [19] methods. In this experiment, we divide the data set into three parts, 60 % for training set, 20 % for testing set, and 20 % for verification set.

The performance of different models on Davis and KIBA dataset is shown in Table 1. It can be seen that the proposed PCNN-DTA achieves

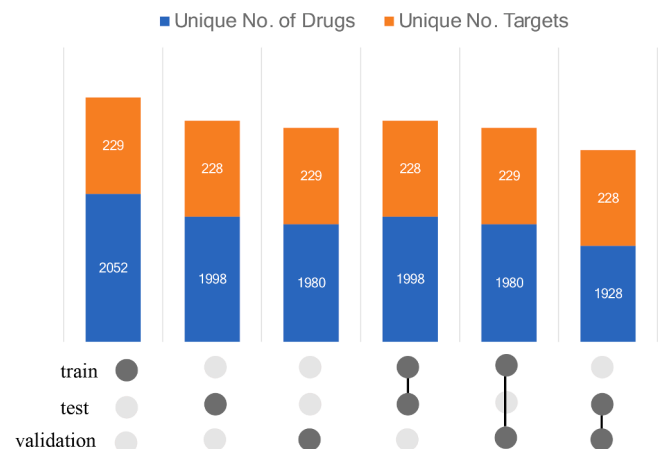


Fig. 3. KIBA random split setting.

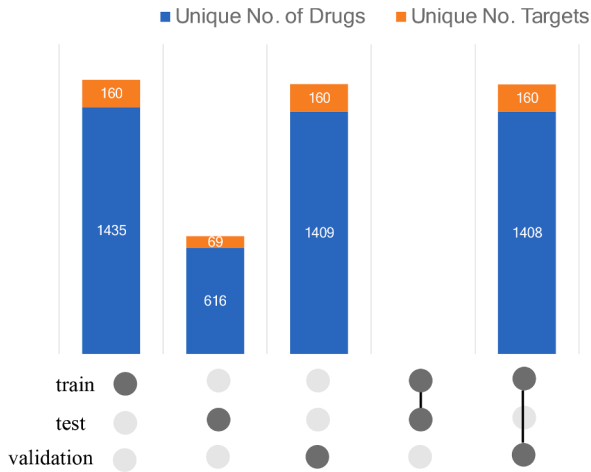


Fig. 4. KIBA cold-start setting.

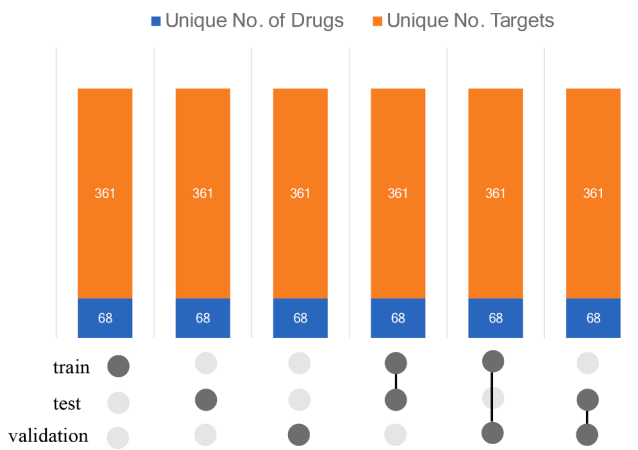


Fig. 5. Davis random split setting.

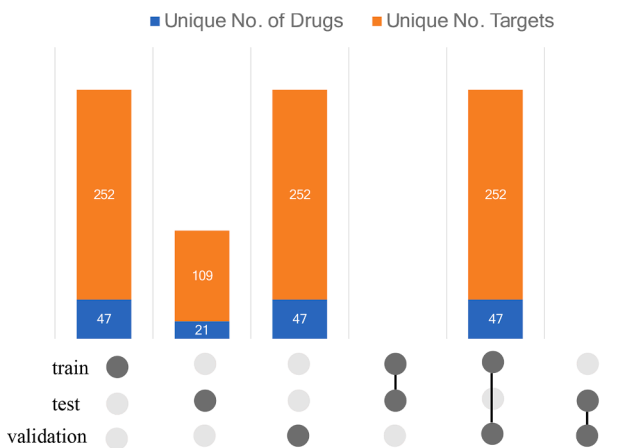


Fig. 6. Davis cold-start setting.

the best results on all evaluation metrics among the existing models PCNN-DTA obtained a 0.0125 higher CI, a 0.031 lower MSE, a 0.026 higher R and a 0.04 higher r_m^2 than the second-best scores achieved on the Davis dataset, and obtained a 0.01351 higher CI, a 0.037 lower MSE, a 0.027 higher R and a 0.072 higher r_m^2 than the second-best scores achieved on the KIBA dataset.

Table 1

Comparison of results with baselines on the Davis and KIBA datasets.

Dataset	Method	CI	MSE	R	r_m^2
Davis	PCNN-DTA	0.8873 (0.003)	0.203 (0.012)	0.828 (0.011)	0.661 (0.0159)
	DeepDTA	0.875 (0.006)	0.239 (0.019)	0.802 (0.008)	0.571 (0.026)
	GraphDTA	0.866 (0.005)	0.240 (0.009)	0.793 (0.003)	0.621 (0.009)
	ML-DTI	0.863 (0.005)	0.234 (0.012)	0.802 (0.009)	0.601 (0.032)
	PCNN-DTA	0.88151 (0.001)	0.151 (0.006)	0.884 (0.003)	0.774 (0.007)
KIBA	DeepDTA	0.868 (0.001)	0.188 (0.002)	0.857 (0.002)	0.697 (0.014)
	GraphDTA	0.838 (0.003)	0.208 (0.005)	0.838 (0.005)	0.696 (0.012)
	ML-DTI	0.861 (0.002)	0.189 (0.003)	0.854 (0.004)	0.702 (0.015)

4.3. The performance of PCNN-DTA of cold-start

We also tested our method under cold start setting. The cold start setting means evaluate model performance on the unseen drugs and targets. The cold-start setting is more realistic and challenge which often used to simulate the new drug and target. In our experiments, we set the training and test sets did not share drugs and targets.

The results for cold-start setting are shown in Table 2. The bold number indicates the best results on the datasets. Compared with the baselines, our model achieves significant improvements on KIBA dataset. As shown, PCNN-DTA gained 0.656 CI index and 0.535 MSE. However, our model is not as competitive on Davis dataset in some metrics due to small size of Davis dataset.

As a result, our model is more likely to perform robustly in unseen drugs and targets application.

5. Discussion

This paper has presented a novel drug-target binding affinity prediction method, PCNN-DTA. A simple framework for building feature pyramids inside CNN is applied to extract meaningful representation for drug and target sequences. Additionally, we applied multi-head cross attention mechanism to fuse the extracted features. Through a series of comparison experiments on two publicly available: KIBA, Davis, the experimental results show that our model performs better than existing models on some evaluation metrics. When cold-start applications, PCNN-DTA proved to be more robust than other compared method in the KIBA dataset. We are planning to design a more powerful pyramid

Table 2

Comparison of results with baselines on the Davis and KIBA datasets under cold start setting.

Dataset	Method	CI	MSE	R	r_m^2
Davis	PCNN-DTA	0.611 (0.034)	0.626 (0.125)	0.191 (0.062)	0.039 (0.024)
	DeepDTA	0.630 (0.036)	0.771 (0.236)	0.270 (0.054)	0.073 (0.029)
	GraphDTA	0.618 (0.030)	0.787 (0.077)	0.235 (0.088)	0.061 (0.047)
	ML-DTI	0.626 (0.038)	0.725 (0.146)	0.246 (0.063)	0.062 (0.028)
	PCNN-DTA	0.656 (0.010)	0.535 (0.041)	0.460 (0.019)	0.194 (0.016)
KIBA	DeepDTA	0.642 (0.007)	0.591 (0.046)	0.453 (0.030)	0.182 (0.024)
	GraphDTA	0.597 (0.014)	0.633 (0.631)	0.369 (0.023)	0.125 (0.016)
	ML-DTI	0.633 (0.015)	0.614 (0.014)	0.412 (0.031)	0.147 (0.022)

feature extractor to different networks, such as Graph Neural Network and Transformer.

CRedit authorship contribution statement

Yuanlong Chen: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Writing – original draft. **Yan Zhu:** . **Zitong Zhang:** Software, Investigation. **Junjie Wang:** Conceptualization, Methodology, Software, Visualization, Investigation. **Chunyu Wang:** Methodology, Formal analysis, Data curation, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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